

Cluster the following points (with (x,y) representing locations) into three clusters.

Data Points	x	y
A1	2	10
A2	2	5
A3	8	4
A4	5	8
A5	7	5
A6	6	4
A7	1	2
A8	4	9

Initial cluster centers: A1, A4, A7

Distance function between two points a(x1,y1) and b(x2,y2):

$$\text{Euclidean distance } d(a,b) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Iteration 1

		(2,10)	(5,8)	(1,2)	
	Point	Dist Mean 1	Dist Mean 2	Dist Mean 3	Cluster
A1	(2,10)	0	3.60	8.06	C ₁
A2	(2,5)	5	4.24	3.16	C ₃
A3	(8,4)	8.48	5	7.28	C ₂
A4	(5,8)	3.60	0	5	C ₂
A5	(7,5)	7.07	3.60	6.70	C ₂
A6	(6,4)	7.21	4.12	5.38	C ₂
A7	(1,2)	8.06	6.40	0	C ₃
A8	(4,9)	2.23	1.41	7.61	C ₂

Point
(2,10)

mean 1
(2,10)

$$d = \sqrt{(2-2)^2 + (10-10)^2} = 0$$

mean 2
(5,8)

$$d = \sqrt{(2-5)^2 + (10-8)^2} = \sqrt{9 + 4} = 3.60$$

mean 3
(1,2)

$$d = \sqrt{(2-1)^2 + (10-2)^2} = \sqrt{1 + 64} = 8.06$$

Point
(2,5)

$$d = \sqrt{(2-2)^2 + (5-10)^2} = 5$$

$$d = \sqrt{(2-5)^2 + (5-8)^2} = 4.24$$

$$d = \sqrt{(2-1)^2 + (5-2)^2} = 3.16$$

		(2,10)	(5,8)	(1,2)	
	Point	Dist Mean 1	Dist Mean 2	Dist Mean 3	Cluster
A1	(2,10)	<u>0</u>	3.66	8.06	c_1
A2	(2,5)	5	4.24	<u>3.16</u>	c_3
A3	(8,4)	8.48	<u>5</u>	7.28	c_2
A4	(5,8)	3.60	<u>0</u>	5	c_2
A5	(7,5)	7.07	3.60	6.70	c_2
A6	(6,4)	7.21	<u>4.12</u>	5.38	c_2
A7	(1,2)	8.06	6.46	<u>0</u>	c_3
A8	(4,9)	2.23	<u>1.41</u>	7.61	c_2

c_1
 (2,10)
 c_2
 (8,4)
 (5,8)
 (7,5)
 (6,4)
 (4,9)
 c_3
 (2,5)
 (1,2)

New Centroids,

Cluster 1, $c_1 \Rightarrow (2, 10)$

Cluster 2, $c_2 \Rightarrow [(8+5+7+6+4)/5, (4+8+5+4+9)/5]$

(6, 6)

Cluster 3, $c_3 \Rightarrow [(2+1)/2, (5+2)/2]$

(1.5, 3.5)

Iteration 2

		(2,10)	(6,6)	(1.5,3.5)	
	Point	Dist Mean 1	Dist Mean 2	Dist Mean 3	Cluster
A1	(2,10)	<u>0</u>	4.47	6.51	c_1
A2	(2,5)	<u>5</u>	4.12	<u>1.58</u>	c_3
A3	(8,4)	8.48	<u>2.82</u>	6.51	c_2
A4	(5,8)	3.60	<u>2.23</u>	5.70	c_2
A5	(7,5)	7.07	<u>1.4</u>	5.70	c_2
A6	(6,4)	7.21	<u>2</u>	4.5	c_2
A7	(1,2)	8.06	<u>6</u>	<u>1.58</u>	c_3
A8	(4,9)	<u>2.23</u>	3.6	6.04	c_1

(2,10)
 (4,9)
 (8,4)
 (5,8)
 (7,5)
 (6,4)
 (2,5)
 (1,2)

Iteration 3

		(3,9.5)	(6.5,5.25)	(1.5,3.5)	
	Point	Dist Mean 1	Dist Mean 2	Dist Mean 3	Cluster
A1	(2,10)	<u>1.11</u>	<u>6.5</u>	<u>6.51</u>	<u>c₁</u>
A2	(2,5)	<u>4.60</u>	<u>4.50</u>	<u>1.58</u>	<u>c₃</u>
A3	(8,4)	<u>7.43</u>	<u>1.95</u>	<u>6.51</u>	<u>c₂</u>
A4	(5,8)	<u>2.06</u>	<u>3.13</u>	<u>5.70</u>	<u>c₁</u>
A5	(7,5)	<u>6.02</u>	<u>0.12</u>	<u>5.70</u>	<u>c₂</u>
A6	(6,4)	<u>6.26</u>	<u>1.34</u>	<u>4.5</u>	<u>c₂</u>
A7	(1,2)	<u>7.76</u>	<u>6.38</u>	<u>1.88</u>	<u>c₃</u>
A8	(4,9)	<u>1.11</u>	<u>4.50</u>	<u>6.04</u>	<u>c₁</u>

(2,10) (8,4) (2,5)
 (5,8) (7,5) (1,2)
 (4,9) (6,4)

Iteration 4

		(3.6,9)	(7,4.3)	(1.5,3.5)	
	Point	Dist Mean 1	Dist Mean 2	Dist Mean 3	Cluster
A1	(2,10)	<u>1.88</u>	<u>6.04</u>	<u>6.51</u>	<u>c₁</u>
A2	(2,5)	<u>4.30</u>	<u>5.04</u>	<u>1.58</u>	<u>c₃</u>
A3	(8,4)	<u>6.66</u>	<u>1.04</u>	<u>6.51</u>	<u>c₂</u>
A4	(5,8)	<u>1.72</u>	<u>4.20</u>	<u>5.70</u>	<u>c₁</u>
A5	(7,5)	<u>5.24</u>	<u>0.7</u>	<u>5.70</u>	<u>c₂</u>
A6	(6,4)	<u>5.54</u>	<u>1.04</u>	<u>4.5</u>	<u>c₂</u>
A7	(1,2)	<u>7.46</u>	<u>6.42</u>	<u>1.58</u>	<u>c₃</u>
A8	(4,9)	<u>0.4</u>	<u>5.57</u>	<u>6.04</u>	<u>c₁</u>

(2,10) (8,4) (2,5)
 (5,8) (7,5) (1,2)
 (4,9) (6,4)

cluster 1 A₁, A₄, A₈

cluster 2 A₃, A₅, A₆

cluster 3 A₂, A₇